A FINITE TYPE MODULE FLAT AT A POINT OF A LOCALLY FINITELY PRESENTED MORPHISM IS FINITELY PRESENTED AT THAT POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The local structure statement with an affine etale neighbourhood. Same chapter flat.tex, label proposition-finite-type-flat-at-point, invoked at proof line 2888 (consumed by S14-algebra-finite-type-flat-at-point at a withdrawn revision and by other cards). Produces an elementary etale neighbourhood over which the pullback of the module is of finite presentation in a neighbourhood of the lifted point.
- Descent of finite presentation along a flat local ring map. Algebra, labels algebra-lemma-descend-properties-modules (proof line 2903) and algebra-lemma-local-flat-ff (proof line 2905). Shows that finite presentation of the pullback at the lifted point descends to finite presentation of the stalk at the given point, since the local ring map is flat hence faithfully flat.

Source reference. Original lemma 03C4: sources/stacks-chow/tags.txt, source lines 4371–4371. Navigation pointer for \ref{algebra-lemma-descend-properties-modules} used inside the counted span (line 2903); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Navigation pointer for \ref{algebra-lemma-local-flat-ff} used inside the counted span (line 2905); the argument that consumes it is retained in the span.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 2888); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism which is locally of finite presentation. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. If $x \in X$ and $\mathcal{F}$ is flat at x over S, then $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$-module of finite presentation.*

Proof. Let $s = f(x)$. By Proposition 05I5 there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ such that the pullback of $\mathcal{F}$ to $X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ is of finite presentation in a neighbourhood of the point $x' \in X_{s'} = X_s$ corresponding to x. The ring map

$$\mathcal{O}_{X,x} \longrightarrow \mathcal{O}_{X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'}),x'} = \mathcal{O}_{X \times_S S',x'}$$

is flat and local as a localization of an étale ring map. Hence $\mathcal{F}_x$ is of finite presentation over $\mathcal{O}_{X,x}$ by descent, see Algebra, Lemma 03C4 (and also that a flat local ring map is faithfully flat, see Algebra, Lemma 00HR). □